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Analysis on building sector's energy consumption and mitigation potential under SSP2

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Abstract

The global energy system will have to face profound transformation in order to realize the 2 degree target. The uncertainty in social and economic development is the one of the most key factors that influences future pathways of energy consumption and carbon emissions. This paper generated energy consumption and carbon emissions scenarios under moderate social economic pathway SSP2 based on a global multi-region energy system model GTIMES, conducted a series of quantitative analysis on different income group regions' building energy consumption and mitigation potential, and made comparisons between SSP2 scenario and RCP2.6 emission pathways. The energy efficiency and energy structure of building sector in low, medium and high income regions will all experience rapid improvement in next decades, however, the mitigation gap between SSP2 scenario and RCP2.6 pathway will still be huge, so more and stronger mitigation measures are needed to be promoted further.

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1. Introduction

The Paris agreement clearly put forward the target of keeping the global average temperature increase to well below 2℃ above pre-industrial level, and pursuing efforts to limit the increase to 1.5℃[1]. In order to make that target realizable, it is necessary to evaluate the energy-use process 's impact on climate change. Among all the end-use sectors, building sector may not always be the largest carbon emitter, but it consumes the most electricity and district heat, the transition in this sector does not only influence its own direct emission, but also have big impacts on emission

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in power sector and heat sector. It is of great importance to study the features of energy consumption and carbon emission in building sector.

The assumption in social and economic development will crucially influence the energy system and further climate change caused by energy use. In order to conduct sensitivity analysis on specific mitigation path and environment policy across different integrated assessment models (IAMs), the socio-economic development assumptions with same features is essential, the Shared Socio-economic Pathways (SSPs) [2-4]which contains 5 trajectories are so put forward. SSP2 stands for moderate development, the world moves slowly towards sustainable target, this trajectory is the easiest to achieve, while major changes of current social, economic and technological trend will not occur, therefore inequity between developed countries and developing countries may still exist with a slight improvement in next decades.

With the imbalanced population growth under SSP2, the energy demand in building sector of developing countries will continue to increase rapidly in next decades. The roadmap for building sector is of highly uncertainty and huge impact, therefore this paper will focus on this sector and analysis the unique characteristic of it.

In order to conduct scenario analysis of building sector's impacts on global energy system and climate change, we plan to design two scenarios based on GTIMES model developed by Tsinghua University: one is the baseline scenario without any energy or environment policies which will be described and analyzed in detail in this paper; the other one is to take 2℃ target into consideration, and combine RCP2.6[5] emission pathway and SSP2 trajectory assumptions, the comparison work will be further conducted. After these two scenarios have been calculated, we will make a comparative analysis of them and find the gap between SSP2 trajectory and 2℃ target in building sector. Based on the comparison work, we can figure out what kind of policy measures may significantly promote the carbon emission reduction within building sector.

2. Methodology

GTIMES model is a dynamic linear programming model on multiple regions, which is developed on the basis of China TIMES by Tsinghua University[6-10]. The whole world is divided into 14 regions in the model according to social development level and geographical distribution, and the energy system is divided into two sections, one is supply side including upstream sector and electricity sector, the other one is demand-side including agriculture, industry, transportation and building sectors. A partial equilibrium in energy system will be reached after the programming of GTIMES model, and the whole computational process is driven by the energy service demand of each region.

Thusly, the prediction of end-use sectors' energy service demand (ESD) based on SSPs is of great importance. Fundamental work of the prediction is to quantize the major socio-economic assumptions of SSP2 trajectory, we choose population, GDP and urbanization rate as core driving factors, and conducted quantitative work based on the research of the United Nations, International Institute for Applied System Analysis, National Center for Atmospheric Research and OECD Environment Directorate, etc.[11-14]. The data of industrial structure and households size are obtained by fitting method based on previous research and related methodology.

For building sector, we considered heating, cooling, lighting, cook and other electrical appliances as basic service and divided the total demand into commercial building and residential building for their differences in functions and drivers. The common influence factors of building sector are GDP, population, household size. The main function of commercial building is to provide service for business activity, so the change of ESD in commercial building is closely linked to the development of service sector, we choose the added value of tertiary industry as one of its most important drivers. As for residential building, its main use is to provide basic services for residents' daily life, this part of energy demand is most influenced by people's income level, so we include GDP per capita in the forecasting work.

Since the development of building sector is closely related to income level, in order to display the model results intuitively and conduct analysis, this paper divide 14 regions into three classes of region according to the income level division method of the World Bank. High-income region includes ACP countries, America, Canada, Japan, Korea and Western Europe; middle-income region includes China, Eastern Europe, the Former Soviet Union countries, Latin America and Middle East; African countries, India, and other countries are classified as low-income region.

3. Results

3.1. Energy service demand

The gross energy service demand in building sector will keep growing in model years. Under the SSP2 scenario assumption that the society, economy and technology will develop smoothly and the society will go slowly to the sustainable development target, the world's total ESD in building sector will be nearly 230 EJ in 2050, 120% higher than in 2010. For commercial building energy sector, demand will grow to 110 EJ in 2050 from 2010's 43 EJ, main growth happens in cooling part, because most developing countries are located in tropical and subtropical regions, with the urbanization process and service industry development in next decades, cooling demand in commercial building will increase rapidly. For residential building, the gross demand is 129EJ, 64 EJ more than in 2010, lighting part is the biggest contributor of ESD growth, mainly for the particularly low lighting demand in low-income countries.

3.2. Final energy consumption in commercial building

Final energy demand growth in commercial building sector mainly happens in low-income and middle-income regions. For high-income region, energy consumption in commercial building grows slowly before 2035 and then remains almost steady, for the service industries in these regions have already been well-developed. However, in low-income and middle-income regions, service industries have broad space to develop based on the general society development path, leading to the rapid growth of commercial building. Thusly, commercial building final energy demand in low-income and middle-income will enter a period of explosive growth, and will be increased by more than 230% and 510% respectively from 2010 to 2050.

Energy intensity in commercial building sector will experience a rapid decrease in all three classes of region, especially in high-income region. With the development and spread of technology, and more importantly the transition to energy-saving and low carbon production mode, energy consumption per unit added value decrease quickly in next 35 years, and reaches 0.38, 0.36,0.33 PJ/billion US dollar in high-income, middle-income, and low-income regions respectively. For most high-income region, the building area has already maintained a high level. take America for instance, its average housing area per capita is about 65-70 m2, almost 2 times of China's level, and their life standards have also reached a relatively high level for now, thusly the gradually shift to energy-saving consuming habits can directly lead to the sharp reduction in energy intensity of commercial building, the energy consumption per unit added value in high-income region will decrease by 47% from 2010 to 2050.

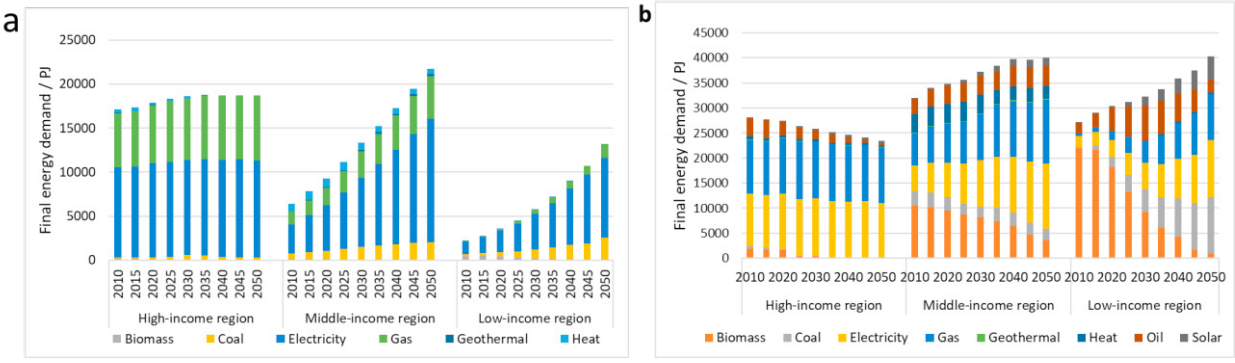


Fig. 1. Final energy demand in (a) commercial building; (b) residential building

3.3. Final energy consumption in residential building

The final energy demand trends of three regions present big differences. In high-income region, as for the per capita floor space has already reached a high level and related service demand does not have a big increasing space, and energy efficiency will be improved in future in the meantime, energy consumption will gradually decrease, reaching 83% of 2010's level in 2050. While for other regions, especially in low-income region, both the growth in population

and improvement of living standard will result in the rise in energy demand, so even with the efficiency improvement and habits changing, there will still be a rising tendency in next 25 years for middle-income region and 35 years or even longer for low-income region. In summary, the change percentage from 2010 to 2050 for high-income, middle-income and low-income regions will be -17%, 25% and 49% separately.

Energy consumption per capita will not experience a significant growth while the living standard of developing countries raises, mainly helped by the technical advancements. This index in high-income region will decrease continuously in next 35 years, because of the saturated demand and efficiency improvement, the average energy consumption per capita will reach 30GJ in 2050, while still slightly higher than Japan's current level. In other regions, the population will continue to grow in the short term, besides, the living standard in most countries is much lower than developed countries and need to be raised, so this index will experience a period of growth, and will still be below 15 GJ and 10 GJ in 2050. The main reason that caused the obvious differences between three classes of regions is the high-standard lifestyle and diversity service demand, which makes the energy consumption in high-income region cannot be decreased rapidly, while in other regions, resident tend to improving their living standard under the concept of energy-saving and environmental-friendly.

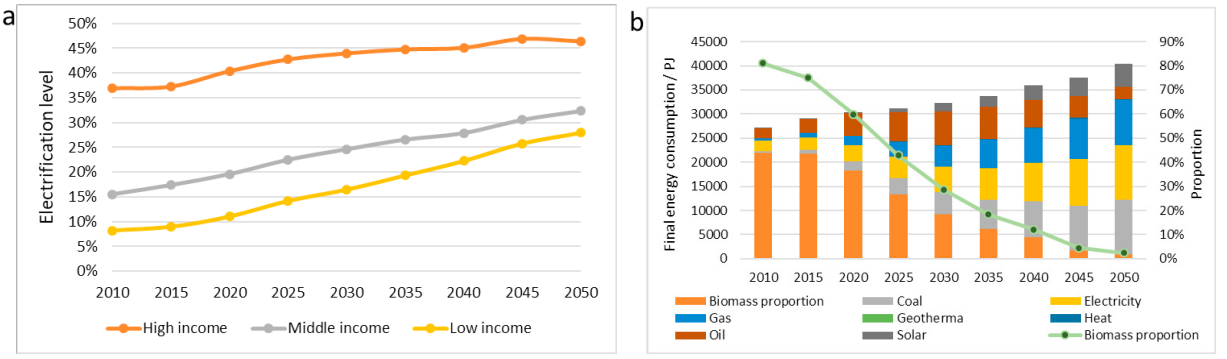


Fig. 2 (a) Residential electrification level; (b) Residential final energy mix in low-income region

Electrification level of residential building will keep raising in a global context, especially in less developed countries. In 2010, electricity only accounted less than 10% in low-income region's residential building energy consumption, mainly because of the lack of related infrastructure such as generation plant and electricity grid, while this proportion will be raised to around 28% in 2050, which will distinctly help to improve the efficiency and cleanness of energy-consumption process. In high-income region, the electrification level of residential building has already reached a relatively high level, and basic needs have been satisfied, the future driver rely mainly on the development of smart cities and powerful electricity grid, the growing trend will continue but not as fast as in low-income and middle-income regions, in 2050, the proportion of electricity in total energy demand will be around 46%, with a 10 percentage point increase from 2010.

The amount of direct-combustion traditional biomass will be reduced quickly in low-income region, from which the environment will significantly benefit. In less developed countries such as some African countries, traditional biomass has been widely used in residential building, accounting for over 80% of residential building energy consumption in 2010, not only the energy efficiency is quite low, but also the indoor air pollution caused by combustion process will cause serious and direct effect on human health. In SSP2 scenario, with the socioeconomic development, even though none specific energy or environment policies are taken, traditional biomass combustion will be replaced by other energy use method gradually, the proportion of biomass will reduce to nearly 2% in 2050.

3.4. Carbon emissions for commercial and residential building

Middle-income region contributes the most in the growth of commercial building sector's CO2 emission, while in residential building sector, over 90% of CO2 emission increment are made in low-income region. Most countries belonged to middle-income region have experienced a period of second industries' rapid development, and will enter the late stage of industrialization, take China as instance, tertiary industry has become one of the most important drive force of economy, thusly, the energy demand and CO2 emission in commercial buildings will grow rapidly in next

decades. However, the most important and urgent task of low-income region is to improve the living standard of the growing population, so the CO2 emission in this region continues to increase greatly until 2050. For high-income region, both tertiary industry and living standard has reached a relatively high level, so the carbon emission will slightly decrease because of the promotion of low-carbon lifestyle and related technology.

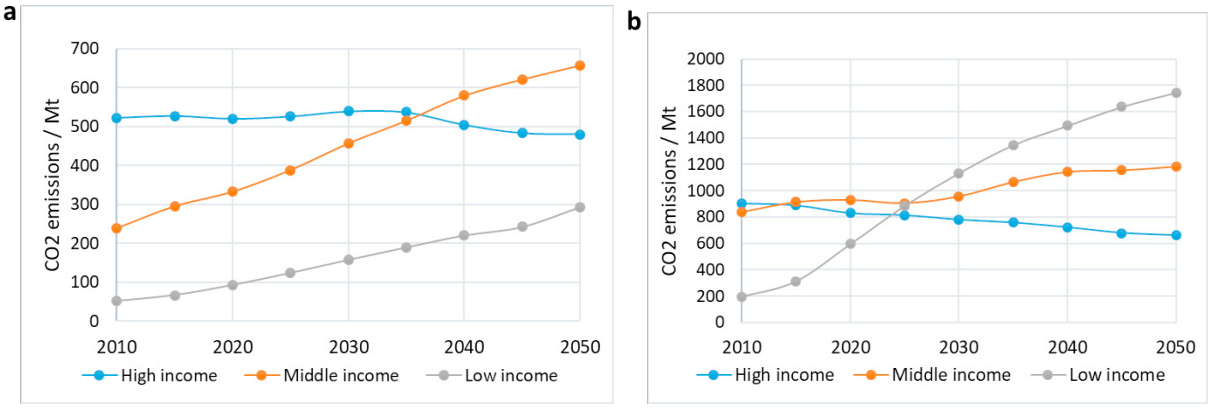


Fig. 3. (a) CO2 emissions in commercial building; (b) CO2 emissions in residential building

3.5. Carbon emission gaps between SSP2 scenario and RCP2.6 scenario

There is a great gap between SSP2 scenario and RCP2.6 scenario, both in commercial building and residential scenario. For commercial building sector, the CO2 emission of RCP2.6 scenario in 2020, 2030, 2050 will be 7.2%, 46% and 80% lower than in SSP2 scenario. And for residential sector, the reduction percentage point is smaller than commercial building, but the amount of emission reduction will reach 2400 Mt in 2050, more than 2 times of commercial building, and the cumulative emission reduction in commercial and residential part will be 15 Gt and 30 Gt respectively from 2010 to 2050.

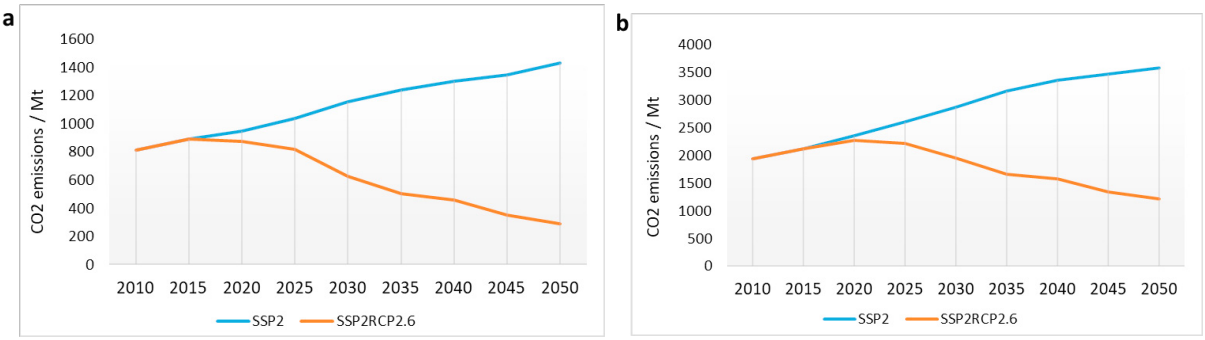


Fig. 4. CO2 emissions gap between SSP2 and RCP2.6 in (a) commercial building; (b) residential building

4. Conclusions and discussions

In SSP2, the growth of final energy in building sector mainly happens in less developed countries, with two main reasons, one is their continuous increase of population and the other is the rapid urbanization process. While in high-income region, the saturation of residential demand and relatively mature development of service industry indicates the driver force will be very weak, so the energy demand will stay constant or even decrease a little with the rapid technical advancement.

Energy consumption per unit added value of commercial building and per capita of residential sector will decrease quickly, especially in high-income region, mainly benefitting from the technical advancement of energy-using equipment such as led light and their widespread application, and the spread of energy-saving and low-carbon concept may also contribute. However, it is worth mentioning that the lifestyle transition may be possibly rather slowly if none resource constraint or environment constraint are considered.

Because of the remaining uneven technological development between regions, the process energy efficiency improvement and carbon mitigation will be of great differences between regions. To be specific, the CO2 emission of high-income region is able to soon peak and the decrease in spite of its uptrend of ESD, while in middle-income and low-income regions, CO2 emission will keep to grow and will not peak before 2050 both in commercial building and residential building.

SSP2 pathway will be needing stringent mitigation measures to meet the 2 degree target. SSPs are mainly focused on socio-economic development, while effort in combating climate change and improving environment are not described in detail, so it would be of great significance to take emission constraints, emission tax and related energy policies into consideration. Under the constraint of RCP2.6, it is obvious that the energy demand of both commercial and residential building sector would decrease with the system cost increases, and the technology development and habits change will also be influenced, related quantitative research based on GTIMES can provide a reference in policymaking of energy system, especially of building sector.

Acknowledgements

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